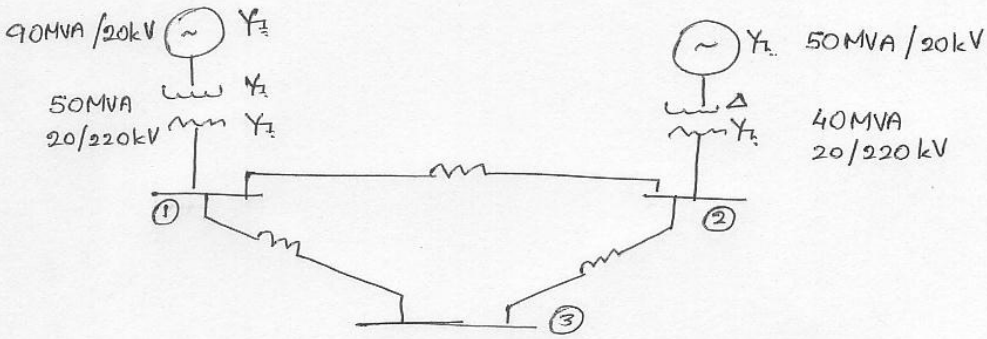
EE 2083- Power Systems II
Assignment I - Fault Analysis
Model Answers.

Question 1



(i) Quantity in per unit = $\dfrac{\text{Actual Quantity}}{\text{Base value of the Quantity}}$

$$Z_{pu} = \frac{Z_{actual}}{Z_{base}}$$

$$Z_{base} = \frac{(V_{base})^2}{S_{base}}$$

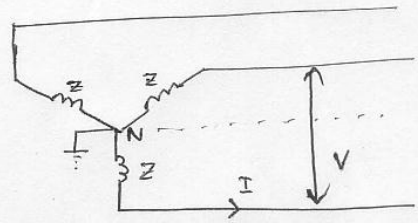
$$Z_{pu} = \frac{Z_{actual}}{(V_{base})^2/S_{base}}$$

$$Z_{pu} = \frac{Z_{actual}}{(V_{base})^2} \cdot S_{base}$$

Note:

$S = \sqrt{3}\,V I$ ← line-to-line voltage, ← line current

$Z = \dfrac{V/\sqrt{3}}{I}$ ← Phase voltage, ← line current



$$Z_{pu,new} = \frac{Z_{actual}}{Z_{base,new}}$$

$$= \frac{(Z_{pu,old})(Z_{base,old})}{(Z_{base,new})}$$

$$= (Z_{pu,old}) \cdot \frac{(V_{base,old})^2/S_{base,old}}{(V_{base,new})^2/S_{base,new}}$$

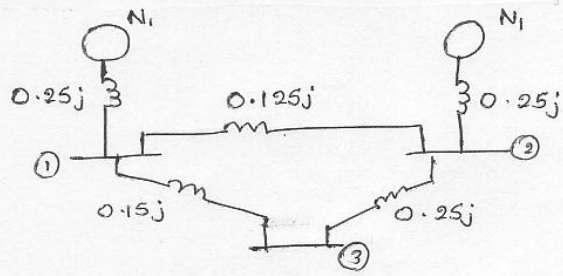
$$= (Z_{pu,old}) \frac{(V_{base,old})^2}{(V_{base,new})^2} \cdot \frac{S_{base,new}}{S_{base,old}}$$

If $V_{base,old} = V_{base,new}$

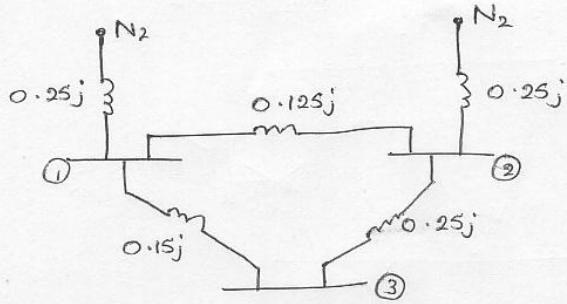
then $Z_{pu,new} = (Z_{pu,old}) \dfrac{S_{base,new}}{S_{base,old}}$

Item	S_{old}	S_{new}	Old, pu			New, pu		
			$X_{1,old}$	$X_{2,old}$	$X_{0,old}$	$X_{1,new}$	$X_{2,new}$	$X_{0,new}$
G1	90	100	0.135	0.135	0.045	0.15	0.15	0.05
G2	50	100	0.075	0.075	0.025	0.15	0.15	0.05
T1	50	100	0.05	0.05	0.05	0.1	0.1	0.1
T2	40	100	0.04	0.04	0.04	0.1	0.1	0.1
L12	100	100	0.125	0.125	0.3	0.125	0.125	0.3
L13	100	100	0.15	0.15	0.35	0.15	0.15	0.35
L23	100	100	0.25	0.25	0.7125	0.25	0.25	0.7125

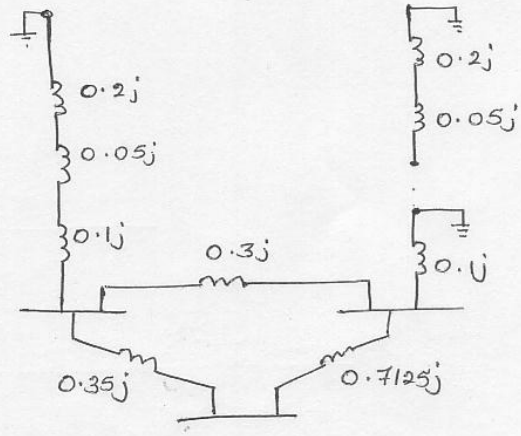
ii) Positive sequece Nelwork



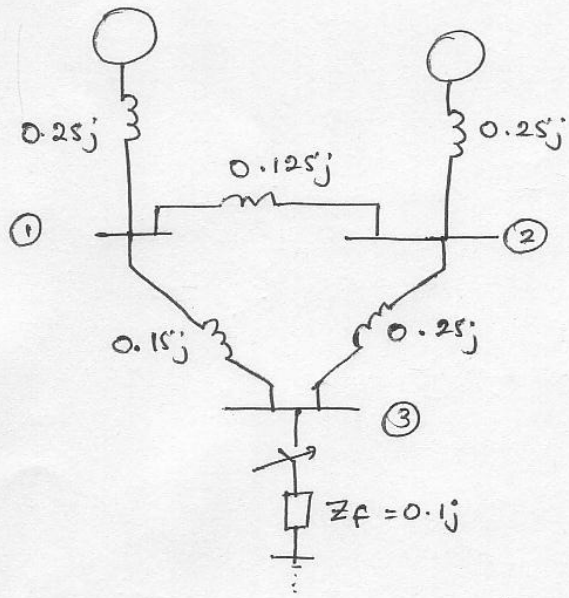
Negative sequence Network

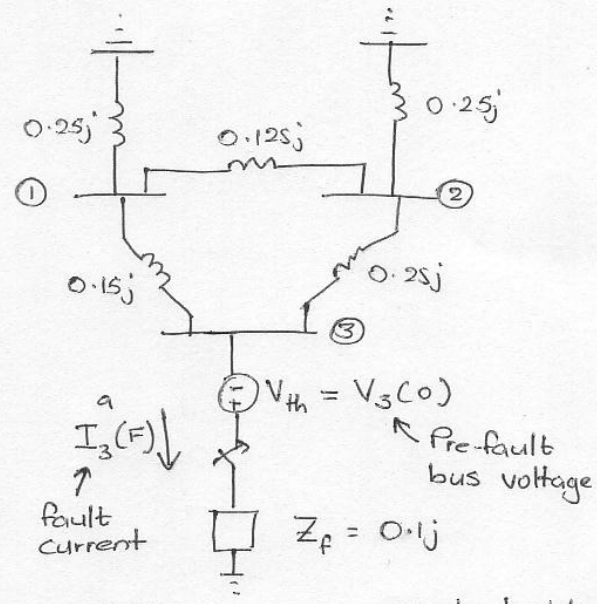


Zero sequence Network



(iii) a) A balanced 3-ϕ fault at bus-3 through a fault impedance $Z_f = 0.1j$ pu.





positive sequence network (per-phase network - Phase a) $\Rightarrow$ Thevenins equivalent network

$$Z_{1s} = \frac{(0.15j)(0.125j)}{0.15j + 0.125j + 0.25j}$$

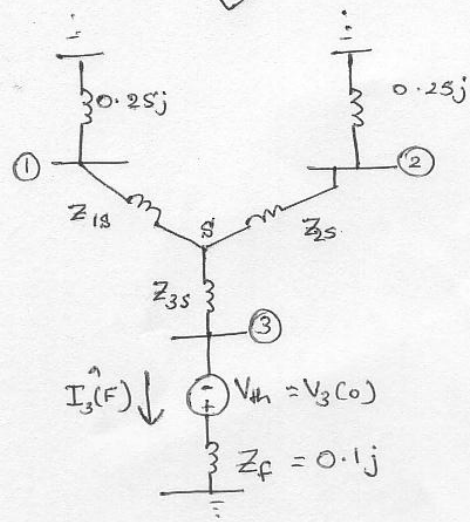
$$= 0.0357143j$$

$$Z_{2s} = \frac{(0.25j)(0.125j)}{0.15j + 0.125j + 0.25j}$$

$$= 0.0595238j$$

$$Z_{3s} = \frac{(0.15j)(0.25j)}{0.15j + 0.125j + 0.25j}$$

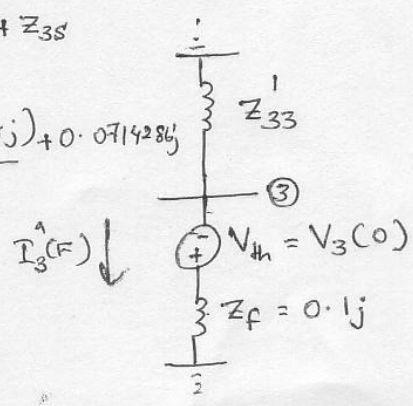
$$= 0.0714286j$$



$$Z'_{33} = \frac{(0.25j + Z_{1s})(0.25j + Z_{2s})}{(0.25j + Z_{1s}) + (0.25j + Z_{2s})} + Z_{3s}$$

$$= \frac{(0.2857143j)(0.3095238j)}{0.5952381j} + 0.0714286j$$

$$= 0.22j$$



Assume : Generators' e.m.f = Rated Values.

$\Rightarrow$ Pre-fault bus voltages = 1pu.

i.e. $V_1(0) = V_2(0) = V_3(0) = 1pu$

$\therefore$ Fault current, $I_3^a(F) = \dfrac{V_3^a(0)}{Z_{33}^1 + Z_f}$

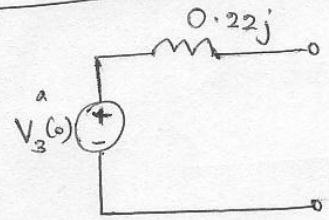
$$= \frac{1}{0.22j + 0.1j} = -3.125j \text{ pu}$$

$$= \underline{\underline{820.1 \angle -90 \text{ A}}}$$

b) A single line-to-ground fault at bus-3 through a fault impedance $Z_f = 0.1j$ pu

Unbalanced fault at bus 3 $\Rightarrow$ Need to draw sequance networks creating thevanin's equivalents as looking from Bus-3

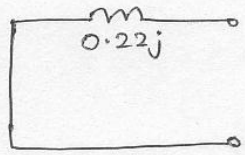
(+)ve sequence [from part (a)]



$Z_{33}^1 = 0.22j$

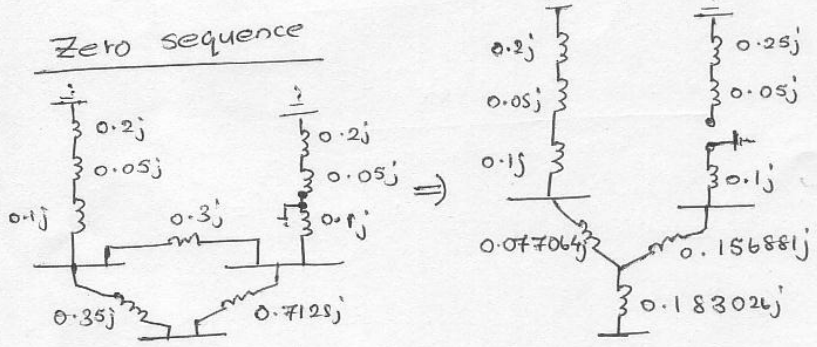
$V_3^1(0) = 1pu$

(−)ve sequence



$Z_{33}^2 = Z_{33}^1 = 0.22j$

Zero sequence



$Z_{33}^0 = 0.34j$

Assume : Fault on phase-a

$$\therefore V_a = Z_f I_a$$

$$I_b = I_c = 0$$

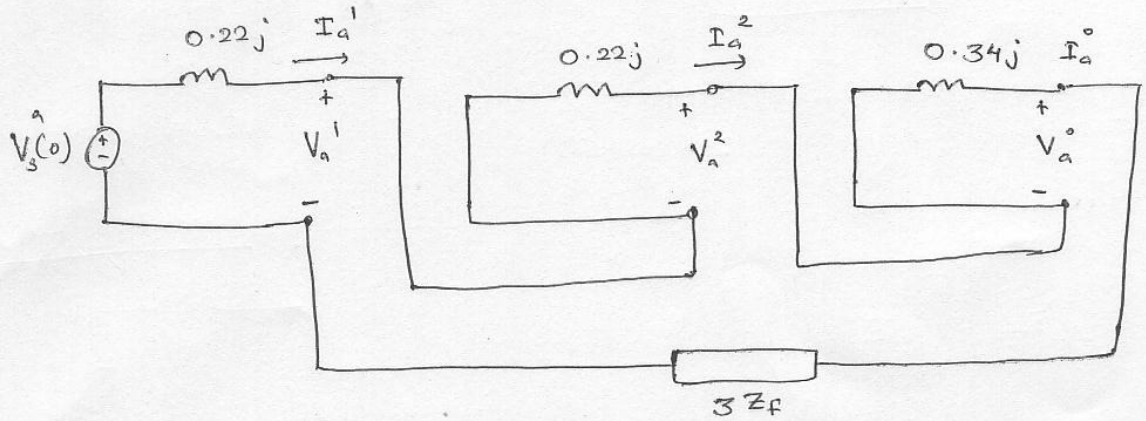
$$\begin{bmatrix} I_a^0 \\ I_a^1 \\ I_a^2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} I_a \\ I_b = 0 \\ I_c = 0 \end{bmatrix}$$

$$I_a^0 = I_a^1 = I_a^2 = \frac{1}{3} I_a$$

Also $$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}\begin{bmatrix} V_a^0 \\ V_a^1 \\ V_a^2 \end{bmatrix}$$

$$\therefore V_a = V_a^0 + V_a^1 + V_a^2 = Z_f I_a$$

$$= Z_f (3 I_a^0) = 3 Z_f I_a^0$$



$$I_a^0 = I_a^1 = I_a^2 = \frac{V_3^a(0)}{Z_{33}^1 + Z_{33}^2 + Z_{33}^0 + 3Z_f}$$

$$= \frac{1}{0.22j + 0.22j + 0.34j + 3(0.1)j}$$

$$= -0.926j \text{ pu}$$

$$I_a = 3 I_a^0 = 3(-0.926j) \text{ pu}$$

$$= -2.78j \text{ pu}$$

$$= \underline{\underline{729.56 \angle -90 \text{ A}}}$$

c) Line-to-line fault at bus-3 through a fault impedance $Z_f = 0.1j$ pu

Assume :- Fault through an impedance $Z_f = 0.1j$ pu between phases b and c.

$$\therefore \quad V_b - V_c = Z_f I_b$$

$$I_c = -I_b \quad \& \quad I_a = 0$$

$$\begin{bmatrix} I_a^0 \\ I_a^1 \\ I_a^2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} 0 \\ I_b \\ -I_b \end{bmatrix}$$

$$\Rightarrow \quad I_a^0 = 0$$

$$\left.\begin{aligned} I_a^1 &= \frac{1}{3}(a - a^2) I_b \\ I_a^2 &= \frac{1}{3}(a^2 - a) I_b \end{aligned}\right\} \Rightarrow I_a^1 = -I_a^2$$

For a 3-ϕ source :

Sequence components

$$V_a^0 = 0 - Z^0 I_a^0$$

$$V_a^1 = E_a - Z^1 I_a^1$$

$$V_a^2 = 0 - Z^2 I_a^2$$

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}\begin{bmatrix} V_a^0 \\ V_a^1 \\ V_a^2 \end{bmatrix}$$

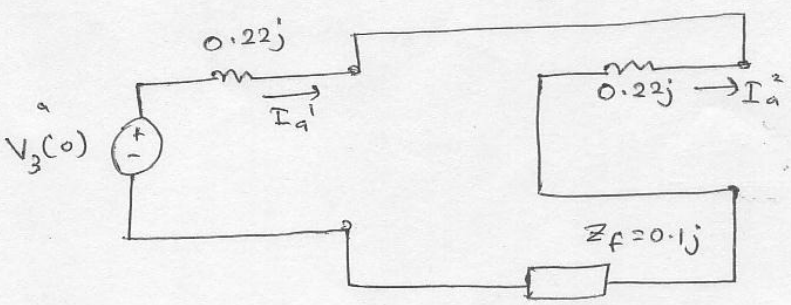
$$\Rightarrow \quad V_b - V_c = (a^2 - a)(V_a^1 - V_a^2)$$

$$Z_f I_b = (a^2 - a)\left[E_a - (Z^1 + Z^2) I_a^1\right]$$

$$Z_f \frac{3(I_a^1)}{(a - a^2)} = (a^2 - a)\left[E_a - (Z^1 + Z^2) I_a^1\right]$$

$$3 Z_f I_a^1 = 3\left[E_a - (Z^1 + Z^2) I_a^1\right]; \quad \because (a - a^2)(a^2 - a) = 3$$

$$\therefore \quad I_a^1 = \frac{E_a}{Z^1 + Z^2 + Z_f}$$



$$I_a^1 = -I_a^2 = \frac{V_3^a(0)}{Z_{33}^1 + Z_{33}^2 + Z_f}$$

$$= \frac{1}{0.22j + 0.22j + 0.1j}$$

$$= -1.8519j \, .$$

At bus 3

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_a^0 \\ I_a^1 \\ I_a^2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} 0 \\ -1.8519j \\ 1.8519j \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ -3.2075 \\ 3.2075 \end{bmatrix} \text{ pu}$$

fault current

$I_c = -I_b$

$= \underline{\underline{841.75A}}$

d) A <u>solid</u> <u>double-line-to-ground</u> fault at <u>bus 3</u>.

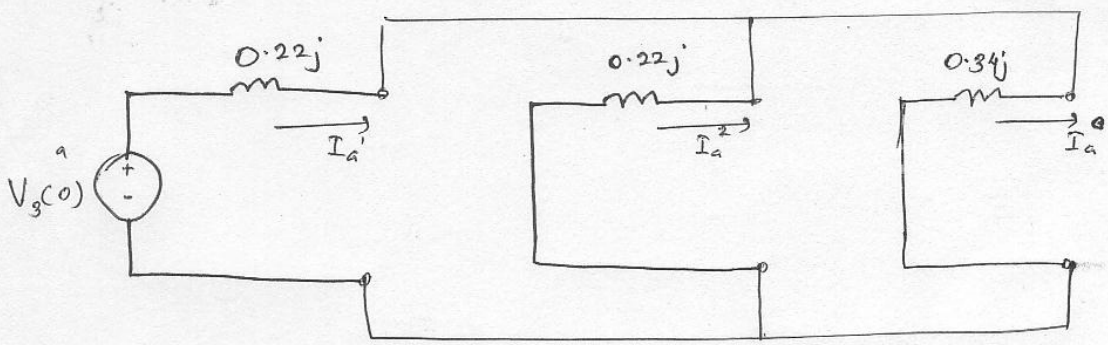
$$I_a = 0 \implies I_a^0 + I_a^1 + I_a^2 = 0$$

$$V_b = V_c = 0$$

Assuming fault between lines b and c.

$$\begin{bmatrix} V_a^0 \\ V_a^1 \\ V_a^2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b = 0 \\ V_c = 0 \end{bmatrix}$$

$$V_a^0 = V_a^1 = V_a^2 = \frac{1}{3} V_a$$



$$I_a^1 = \frac{V_3^a(0)}{Z_{33}^1 + \left(\dfrac{Z_{33}^2 \, Z_{33}^0}{Z_{33}^2 + Z_{33}^0}\right)}$$

$$I_a^1 = \frac{1}{0.22j + \left(\dfrac{0.22j \cdot 0.34j}{0.22j + 0.34j}\right)}$$

$$I_a^1 = -2.828j$$

$$I_a^2 = -\left[-2.828j\left(\frac{0.34j}{0.22j + 0.34j}\right)\right] = 1.717j$$

$$\therefore \quad I_a^0 = 1.111j$$

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_a^0 = 1.111j \\ I_a^1 = -2.828j \\ I_a^2 = 1.717j \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ -3.93597 + 1.666j \\ 3.93597 + 1.666j \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 4.274 \angle 157.05^\circ \\ 4.274 \angle 22.75^\circ \end{bmatrix}$$

fault current $= I_b + I_c$

$= \underline{\underline{3.332 \angle 90^\circ \text{ pu}}}$

$= \underline{\underline{874.42 \angle 90^\circ \text{ A}}}$